Day-10 Oct-12

IPv4 Header

Data → Do Data

Data + L4 Header → Segment

Data + L4 Header + L3 Header → Packet

L2 Trailer + Data + L4 Header + L3 Header + L2 Header → Frame

Protocol Data Units (PDUs)

IPv4 Header.

Offsets	Octet	0								1								2								3			
Octet	Bit	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
0	0	Version				IHL				DSCP						ECN		Total Length													
4	32	Identification																Flags			Fragment offset									
8	64	Time To Live								Protocol								Header Checksum													
12	96	Source IP Address																													
16	128	Destination IP Address																													
20	160	Options (if IHL>5)																													
24	192																														
28	224																														
32	256																														

Read this chart L→R & T→B

Version.

It's length is 4 bit. It identifies the IP in use. IPv4 or IPv6

- IPv4 = 4 (0100)
- IPv4 = 6 (0110)

What about IPv5? Just for curiosity. No need for CCNA

There was experimental protocol called Internet Stream Protocol which used IP value of 5. but it was never used publically. So, we can say that's the last IPv5.

The IPv6 header has diffent structure than IPv4 header.

IHL (Internet Header Length).

- The final field of IPv4 header i.e Options is variable in length, so this field is necessary to indicate the total length of header.
- It Here's identifies the length of the header in 4-byte increments.
  for eg If value in this field is 5.
  value 5= 5x4 = 20 bytes
  This means length of header is 20 bytes.
  Min value in this field = 5 (= 20 bytes ie when there is no option)
  Max value " " " = 15 (= 60 byte when length of option is max i.e 40 bytes).

4

60 bytes is max length of IPv4 header.

DSCP (Differentiated Services Code Point)

- The field length is 6 bits
- This field is used for QOS (Quality of Service).
  (QOS will be taught later too)
- It is used to prioritize delay-sensitive data, such as streaming voice & video etc
- Eg- If we are loading webpage & internet get slow, no big deal
  If we're having video call & there's a terrible delay & or audio or video freezes, it can totally ruin experience
- This field is used to identify which the traffic should receive priority experience.

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ECN (Explicit Congestion Notification)

- It has length of 2 bits.
- It provides end-to-end (between two ends) notifications of network congestion without dropping packets.

(Normally चाहिँदै congestion हुँदा हुने network मा congestion indicate गर्न Packet drop गरिन्छ। But, ECN ले चाहिँदै Packet drop नगरी notify गर्छ)

- It is an optional feature that requires both endpoints, as well as the underlying network infrastructure, to support it.

Total Length

- It has field length of 16 bits or 2 bytes.
- Indicates total length of packet i.e (IP header + TCP segment + Data)
- It is different than IHL which indicates length of IPv4 ~~IHL~~ (~~Only~~ header only.
- It indicates length in bytes (no 4-byte increments like IHL) Eg: If there is value of 20 in this field it means 20 Byte is the size of packet.

Minimum size of packet = 20 bytes (= IPv4 header with no encapsulated data)

Maximum value of 65, packet = 65,535 bytes (as 15 in 16 bit value)

Identification field

- It has field length of 16 bits or 2 bytes.
- If a packet is fragmented due to being too large, this field is used to identify which packet the fragment belongs to.
- All fragments of the same packet will have their own IPv4 header with the same value in this field. (so that they can be reassembled later)
- Packets are fragmented if larger than the MTU (Max. Transmission Unit)
- The MTU = 1500 bytes. (the max payload size of ethernet frame)
- Fragments are reassembled by receiving host.

Flags.

- It has field length of 3 bits.
- It is used to control/identify fragments.
- It is of 3 bit which fix as follows:

- Bit 0: First bit; reserved, always set to 0.
- Bit 1: Don't Fragment (DF bit), used to indicate a packet that shouldn't be fragmented
- Bit 2: More fragment (MF bit). set to 1 if there are more fragments in packet and, set to 0 for last fragment

Note: If packet is whole (unfragmented) they'll always have their MF bit set to 0. Bcs there're no more fragments left.

Fragment offset field

- It has field length of 13 bits.
- Used to indicate the position of fragment within original, unfragmented IP packet
- Allows fragmented packets to be reassembled even if fragments arrive out of order, as this field lets the receiver know original order of fragment.

Time to live (TTL)

- It has field length of 8 bits
- A router will drop a packet with TTL of 0.
- It is used to prevent infinite loops.
  (If a poor routing configuration, causes a packet to be continually sent around in a loop, never reaching its intended destination, if enough traffic like that accumulates, it could causes network congestion & eventually failure).
- Originally designed to indicate packet's max lifetime.
- It actually indicates ~~hopecount~~ 'hop count': each time the packet arrives at a router, the router decreases the TTL by 1.
- The recommended default TTL is 64.

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Protocol

- It has field length of 8 bits
- Indicates the protocol of encapsulated L4PDU. (layer 4 PDU).
  - Value of 6: TCP
  - Value of 17: UDP
  - Value of 1: ICMP
  - Value of 89: OSPF (dynamic routing protocol)
    - OSPF: Open Shortest Path First

OSPF allows routers to learn routes to distration from their neighbours without us having to manually configure them.

Header Checksum

- It is 16 bit in length.
- A calculated checksum used to check for errors in the IPv4 header (not in Packet only in header)

- When a router receives a packet, it calculates the checksum of the header & compares it to one in this field of header.
- If they donot match, router drops the packet.

- IP relies in encapsulated protocol to detect error in encapsulate data.
- Both TCP & UDP have their own checksum fields to detect errors in encapsulated data.

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Source / Destination IP Address.

- They are 32 bits (4 bytes) in length each. as it is the length of IPv4 address.

- Source IP Address = IPv4 address of sender of packet.
- Destination IP Address = IPv4 address of intended receiver of packet

Options field.

- This field is optional.
- 0 byte - if not used 40 byte - if completely used.
- Rarely used.
- If IHL > 5, it means options are present (this field is used)

We can read chart from wikipedia about struct of options field, but ain't important for CCNA.

QUIZ

1) What is the fixed binary value of first field of IPv4 header?
a) 0010 b) 0110 c) 0001 d) 0100

2) Whic field will cause packet to be dropped if it has value of 0?
a) TTL b) DSCP c) IHL d) ECN.

3) How are errors in IPv4 packets encapsulated data detected?
a) IPv4 Header Checksum field check for errors
b) The encapsulated protocol (TCP, UDP) checks for errors.
c) Errors in encapsulated data can't be detected.

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